

REEV – Plug-In Hybrid electric vehicles

US/CA markets

Preliminary
March 2025

Final emission targets will be published after
alignment with EPA and CARB





PRELIMINARY INFORMATION

The information included in this presentation is subject to change pending final vehicle specification, vehicle performance during each test cycle, qualifying exemptions, and alignment with EPA and CARB.





DEFINITION OF HYBRID VEHICLES

- ❖ *Hybrid or Hybrid vehicle (HV)* means a vehicle that includes energy storage features (other than a conventional battery system or conventional flywheel) in addition to an internal combustion engine or other engine using consumable chemical fuel. Supplemental electrical batteries and hydraulic accumulators are examples of hybrid energy storage systems. Note that certain provisions in this part treat hybrid vehicles that include regenerative braking different than those that do not include regenerative braking.
- ❖ *Hybrid electric vehicle (HEV)* means a motor vehicle which draws propulsion energy from onboard sources of stored energy that are both an internal combustion engine or heat engine using consumable fuel, and a rechargeable energy storage system such as a battery, capacitor, hydraulic accumulator, or flywheel. This includes plug-in hybrid electric vehicles
- ❖ *Plug-in hybrid electric vehicle (PHEV)* means a hybrid electric vehicle that has the capability to charge the battery from an off-vehicle electric source, such that the off-vehicle source cannot be connected to the vehicle while the vehicle is in motion.

➤ [VF7/VF MPV9 REEV are PHEV](#)

[Source : eCFR :: 49 CFR 535.4 -- Definitions.](#)

US - The regulations are listed in 40CFR86, mostly in Subpart S (86.1801-12 - 86.1870-12):

<https://www.ecfr.gov/current/title-40/chapter-I/subchapter-C/part-86/subpart-S?toc=1>

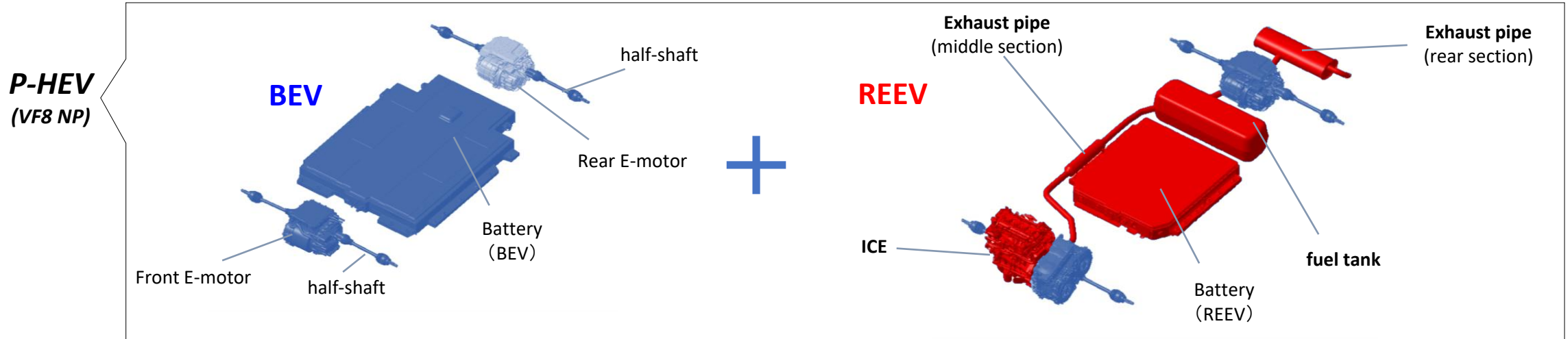
CARB - MY2026 through MY2035 CARB regulations are known as LEV IV.

<https://ww2.arb.ca.gov/our-work/programs/advanced-clean-cars-program/lev-program/low-emission-vehicle-regulations-test>

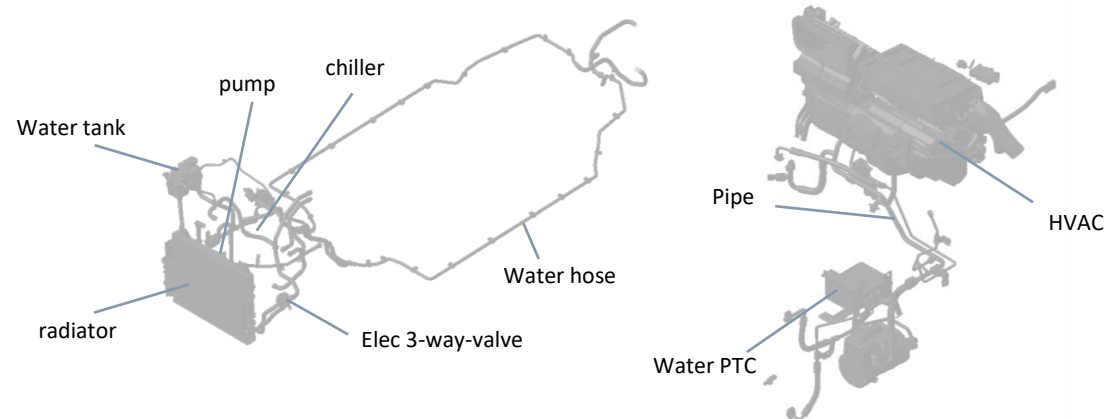


DEFINITION OF PHEV IN SERIES (REEV)

In a series hybrid powertrain, the internal combustion engine is not providing torque directly to the drive wheels. Instead, the engine is powering an electrical generator which provides electrical energy to the traction electric motor.



Improvement at Thermal system (Cooling layout is still under discussion)





EMISSION STANDARDS

	Items	Reg.	Limits	Test method/ procedures	Link
1	Exhaust emissions	40CFR Part 86 Subpart S CARB - LEV IV	LDV NMOG+ NO_x (mg/mile)^b15 PM (mg/mile)^c0.5 CO (g/mile)^d1.7 Formaldehyde (mg/mile)^e4	FTP/HFET/US06/SC03/20FTP Plus 50FTP, Partial soak, quick drive-away, high powered cold start for CARB. Limits shown are fully phased in EPATier4 standards. Fleet average NMOG+NOX targets will determine final BIN selection	§ 86.1811-27 DF: § 86.1823-08 f.1
2	Particulate emissions		1 & 2 mg/mi	FTP, US06	
3	Idle Emissions		-	idle and 2500 rpm	
5	Evaporative Emissions		0.3/0.65gr (low/High altitude) 0.05gr (low altitude) 0.2 gr/gallon (Low altitude) 1.0 gr	2-day/3-day Running Loss ORVR Spitback	§ 86.1813-17
6	Durability Demonstration	40CFR Part 600	-	150,000-mile basis	§ 86.1823-08 § 86.1824-08 § 86.1825-08
7	On-Board Diagnostics and IUPRm		-	Full useful life and limit catalysts, OBD Monitoring tests	§ 86.1806-27
8	CO2 emission and fuel consumption	40CFR Part 600	2027 - Compression-ignition: 0.0348 × WF + 268 (g/mile)	FTP/HFET/GHG/FE Label/Monroney label	40CFR Part600
9	Fuel property	40CFR Part 80	-	E10(10%ethanol) /E0 specification	§ 1065.710(b) 40CFR Part 80
10	Test group	40CFR Part 86	TBD		§ 86.1827-01
11	Coastdown	SAE J1634:2021	VF target (based on VN FL): > 350 km Pure EV > 1000 km mix	SAE J1634:2021 procedure	40CFR Part600.116-12

Final emission targets will be published after alignment with EPA and CARB





EMISSION STANDARDS

Vehicles must meet the emission standards for their full useful life, which is typically 120,000 miles for EPA and 150,000 miles for CARB.

There are two ways to test the vehicles to demonstrate that they meet the emission standards at full useful life:

1 - The manufacturer accumulates the full useful life mileage on test vehicles (~150,000 miles on each vehicle) in accordance with the SRC cycle and then tests the vehicles for exhaust and evaporative emissions compliance. The drawback to this approach is that the vehicles must start the mileage accumulation with frozen (Production intent SW and HD) calibrations, which usually adds time to the certification process.

2- Early in the program, the manufacturer accumulates full useful life mileage on test vehicles (Known as Durability Data Vehicles or DDVs) with a representative, but unfrozen (Not production intent, but representative), calibration. The vehicles are then tested for exhaust and evaporative emissions every 5,000 or 10,000 miles until the last tests at 150,000 miles. The manufacturer then draws a graph and determines how much each emission constituent (NMOG, CO, PM, NOX, etc.) deteriorated from 4000 miles to 150,000 miles. For example, if the CO was 1 g/mi at 4000 miles and 1.2 g/mi at 150,000 miles, the CO deteriorated by 20% from 4k to 150k miles. This results in a multiplicative CO emission Deterioration Factor of 1.2 (CO DF of 1.2). After that, and once the emission calibration is frozen, the manufacturer would then accumulate only 4K miles on a test vehicle (Known as Emission Data Vehicle or EDV) and test it for emissions. The CO result would then be multiplied by 1.2 to determine what the expected CO emissions would be at 150,000 miles. The CO result at 4K miles, after being multiplied by 1.2, is then compared to the CO emission standard that the vehicle must meet at full useful life. The advantage of this approach is that, once the calibration is frozen, the manufacturer only needs to accumulate 4K miles on test vehicles, instead of the 150k miles, to test vehicles for certification. The drawback is that it costs a lot more to launch the initial vehicles (DDVs) to determine DFs.

Because the DDV process takes a long time and is very expensive, EPA decided to offer small volume manufacturers the option of using DFs developed by EPA, known as EPA Assigned DFs. For example, EPA allows the SVM manufacturer to test an EDV at 4k miles and multiply the CO result by 1.1, the CO DF assigned by EPA, to demonstrate compliance with the CO standard. Large volume manufacturers do not have this option and must develop their own DFs by testing DDVs.

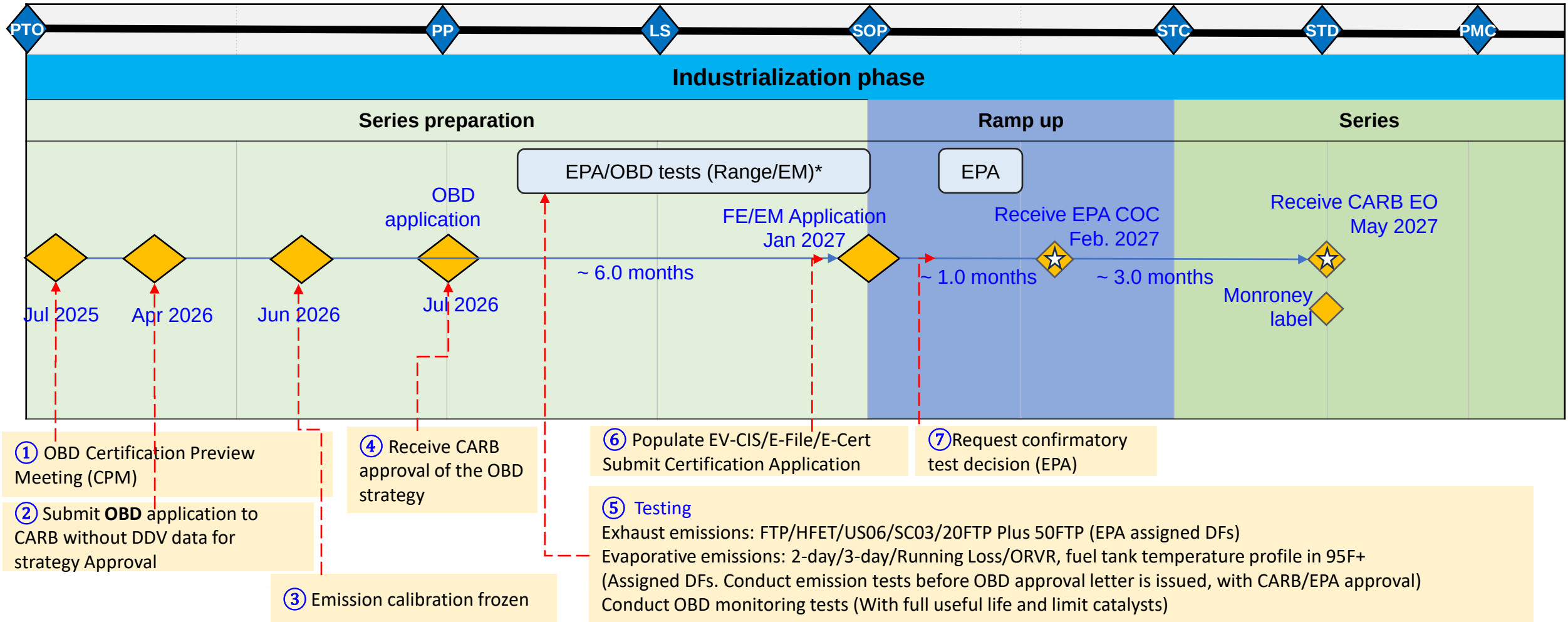
There are also other options for DDV procedures. For example, a manufacturer may age certain emission components (Catalysts, oxygen sensors, etc.) to 150K miles equivalent on a bench instead of on a vehicle and install them on a 4K EDV for full useful life emission tests. These bench procedures must meet certain criteria and be approved by EPA.





HEV'S HOMOLOGATION PROCESS – GENERAL SCHEDULE

The certification test procedures, EV-CIS and E-Cert entries, and OBD strategy approval and associated tests take significantly longer for an ICE or Hybrid model compared to a BEV. The whole process can take about 8 months +/- 2 months. The cost is also significantly higher.





EPA Regulations



US - The regulations are listed in 40CFR86, mostly in Subpart S (86.1801-12 - 86.1870-12): < [eCFR :: 40 CFR Part 86 Subpart S](#) >

- ❖ On March 20, 2024, EPA announced new, more protective final standards to further reduce harmful air pollutant emissions from light-duty and medium-duty vehicles starting with model year 2027.
- ❖ These standards will phase in over model years 2027 through 2032.

EPA tests to ensure that REEV vehicle comply with emission standards:

- Federal Test Procedure (FTP):** Emissions under driving cycle that simulates urban driving conditions
- Highway Fuel Economy Test (HFET):** Emissions and fuel economy during highway driving conditions
- US06 test:** Emissions under aggressive, high-speed driving conditions
- SC03 test:** Emissions with the air conditioning system operating
- Evaporative/Refueling emissions test:** Measures the amount of fuel vapor emissions from the vehicle
- On-Board Diagnostics (OBD) test:** Ensure that OBD system can detect and report emission-related malfunctions

As **PHEVs** (Plug-in hybrid electric vehicle)

[eCFR :: 40 CFR 86.1811-27 -- Criteria exhaust emission standards.](#)

The driving cycle . . .	is identified in . . .
(A) FTP	40 CFR 1066.801(c)(1).
(B) US06	40 CFR 1066.801(c)(2).
(C) SC03	40 CFR 1066.801(c)(3).
(D) HFET	40 CFR 1066.801(c)(5).
(E) ACC II—Mid-temperature intermediate soak	40 CFR 1066.801(c)(8).
(F) ACC II—Early driveaway	40 CFR 1066.801(c)(9).
(G) ACC II High-load PHEV engine starts	40 CFR 1066.801(c)(10).

Fully phased-in Tier 4

LDV	NMOG+ NO _x	(mg/mile) ^b	15
	PM	(mg/mile) ^c	0.5
	CO	(g/mile) ^d	1.7
	Formaldehyde	(mg/mile) ^e	4

Tier 4 NMOG+NO_x Bin Standards for Light-Duty Program Vehicles [mg/mile]

Emission standard is based on a fleet average for a given model year

	FTP, US06, SC03, HFET	ACC II—Mid- temperature intermediate soak (3-12 hours)	ACC II—Mid- temperature intermediate soak (40min)	ACC II—Mid- temperature intermediate soak (10 min)	ACC II— Early driveaway ^b	ACC II— High-power PHEV engine starts ^{bc}
FEL name						
Bin 20	20	20	15	10	32	67

FEL – Family emission limit



CARB Regulations



PHEV (all-electric range of ~ 200km)

If VinFast wishes for the PHEVs to count towards the ZEV program, the REEVs must meet the requirements in 13 CCR 1962.4, see attachment.

ADDITIONAL REQUIREMENTS FOR PHEVs (Requirements in the next slide also apply)

§ 1962.4. Zero-Emission Vehicle Requirements for 2026 and Subsequent Model Year Passenger Cars and Light-Duty Trucks.

<< [View Document - California Code of Regulations](#)>>

- (1) **Certification Range Value:** *Minimum certification range value greater than or equal to 200 miles,*
- (2) **ZEV Durability Requirement for Useful Life:** *For a useful life of 10 years or 150000 miles, which ever comes first, at-least 70% vehicles shall have at-least 70% of certified electric range.*
- (3) **Battery Labeling Requirements** - [1962.6](#)
- (4) **Data Standardization** - [1962.5](#)
- (5) **Service Information Requirements** - [1962.7](#)
- (6) **ZEV Warranty Requirements** - [1962.8](#)
- (7) **Charging Requirements** - [1962.3](#)

If VinFast does not want to count for ZEV program, the minimum requirements mentioned in the standard 13 CCR 1961.4 shall still be met.





HEV

Note that **1961.4 references 1962.4** as well and VinFast must still meet minimum requirements in 1962.4 as described in 1961.4.

§ 1961.4. Exhaust Emission Standards and Test Procedures--2026 and Subsequent Model Year Passenger Cars, Light-Duty Trucks, and Medium-Duty Vehicles.<< [View Document - California Code of Regulations](#) >>

CARB tests to ensure that REEV vehicle comply with standards set for 2026 and subsequent model years:

1. **FTP 50 degree F:** Emission standards at 50°F
2. **FTP 20 degree F - Cold CO standard:** Carbon monoxide emissions under cold conditions
3. **US06 NMOG+NOx and CO standards:** Non-methane organic gases (NMOG) and nitrogen oxides (NOx) emissions
4. **US06 PM standard:** Particulate matter (PM) emissions
5. **SC03 standard:** Exhaust emissions of NMOG+NOx and CO
6. **Highway standard:** Emissions during highway driving
7. **Partial soak:** Tests at 10 minutes, 40 minutes, and 3-6 hours with in between tests for NMOG+NOX
8. **Quick Drive-Away:** Test at low altitude for NMOG+NOX





CARB LOW EMISSION VEHICLE IV (LEV IV) STANDARDS

- The **LEV IV** emission standards were finalized by CARB in August 2022 and approved by the Office of Administrative Law (OAL) in November 2022.
- They are phased-in over the 2026-2030 model years. Manufacturers can certify vehicles to the LEV IV standards in model year 2025 but beginning with model year 2026, all vehicles must be certified to LEV IV standards.
- Changes from LEV III :
 - phase out the use of ZEVs and emission-adjusted PHEVs in the calculation of the combined NMOG+NOx fleet average requirement by the 2029 model year,
 - introduction of new bins including low emission bins and removal of high emission bins
 - introduction of high altitude NMOG+NOx limits, partial soak standards and quick drive-away standards,
 - introduction of fleet standards for medium-duty vehicles.
 - Introduction of new test standards
- 2027+ Model Year certification Requirement: Based on the **0.030 Fleet Average NMOG+NOX** standard, the vehicle must be certified to **SULEV30**, or stricter, category unless certified as a small volume manufacturer





CARB : ADDITIONAL TESTING REQUIREMENTS FOR LEV IV VEHICLES

Test	All PCs / LDTs ≤ 8500 lbs GVW/ All MDPVs
Partial soak NMOG+Nox	Full compliance by 2028 – SULEV30 10-min soak: 0.015 g/mi 40-min soak: 0.023 g/mi 3 – 12-hour soak: 0.030 g/mi
Quick drive-away NMOG+Nox	Full compliance by 2028 – SULEV30: NMOG+NOx: 0.042 g/mi
50°F (10°C) NMOG+NOx and HCHO	Applies at 4,000 miles – SULEV30 NMOG+NOx: 0.060 g/mi HCHO: 0.008 g/mi
Cold CO	Applies at 50,000 miles CO: 10 g/mi
US06 NMOG+NOx, CO and PM	Full compliance of NMOG+NOx and CO by 2028; and 2027 must meet Interim standards; PM full compliance by 2030; Vehicles not meeting final 3 mg/mi PM limit in 2026 – 2029 must meet interim 6 mg/mi limit; FUL
High power cold start US06 PHEV NMOG+Nox	150,000 mile PHEVs SULEV30 NMOG+NOx (g/mi) 2026 to 2028: 0.150 NMOG+NOx (g/mi) 2009 ~ : 0.100
SC03 NMOG+NOx and CO	NMOG+Nox (Low/high altitude) : 0.030/0.050 g/mi CO: 1.0g/mi
HWFET NMOG+Nox	FUL

LEV regulation	Vehicle emission category	Low altitude NMOG + NO _x (g/mile)	High altitude NMOG + NO _x (g/mile)	CO (g/mile)	HCHO (g/mile)	PM ^c (g/mile)
LEV III	LEV160 ^a	0.160	0.160	4.2	0.004	0.003
	ULEV125 ^a	0.125	0.125	2.1	0.004	0.003
	ULEV70	0.070	0.105	1.7	0.004	0.003
	ULEV50	0.050	0.070	1.7	0.004	0.003
	SULEV30	0.030	0.060	1.0	0.004	0.003
LEV IV	SULEV20	0.020	0.050	1.0	0.004	0.003
	ULEV125	0.125	0.160	2.1	0.004	0.001
	ULEV70	0.070	0.105	1.7	0.004	0.001
	ULEV60 ^b	0.060	0.090	1.7	0.004	0.001
	ULEV50	0.050	0.070	1.7	0.004	0.001
	ULEV40 ^b	0.040	0.060	1.7	0.004	0.001
	SULEV30	0.030	0.050	1.0	0.004	0.001
	SULEV25 ^b	0.025	0.050	1.0	0.004	0.001
	SULEV20	0.020	0.030	1.0	0.004	0.001
	SULEV15 ^b	0.015	0.030	1.0	0.004	0.001

^a LEV160 category is removed from LEV IV and ULEV125 is applicable for MYs 2026–2028 for manufacturers other than small volume manufacturers (SVMs). For a SVM, it is available until 2034.
^b ULEV60, ULEV40, ULEV25, and ULEV15 categories are added in LEV IV
^c LEV III and LEV IV both have a phase-in schedule for PM emission standards until they reach the numbers shown in the table

^a Full US06 for MDVs with hp/GVWR > 0.024; US06 Bag 2 for MDVs with hp/GVWR ≤ 0.024

^b Gross Combined Weight Rating, includes the vehicle and its trailer



CARB : FLEET EMISSION REQUIREMENTS

- ❖ Fleet NMOG+NOx requirements for PCs, LDTs and MDPVs remain at 0.030 g/mi from 2026 onward.
- ❖ However, the percentage of a manufacturer's total ZEVs + emission-adjusted PHEVs allowed in the fleet calculation decreases to 0% by 2029, Table (see below).
- ❖ Emission-adjusted PHEVs use a PHEV NMOG+NOx contribution factor that accounts for the all-electric driving range over the FTP and US06 cycles.

Fleet average requirements for PCs, LDTs and MDPVs(150,000 mile durability vehicle basis)		
Model Year	NMOG+NOx, g/mi	Maximum Percent ZEVs + emission-adjusted PHEVs ^a
2025 ^b	0.030	100%
2026		60%
2027		30%
2028		15%
2029+		0%

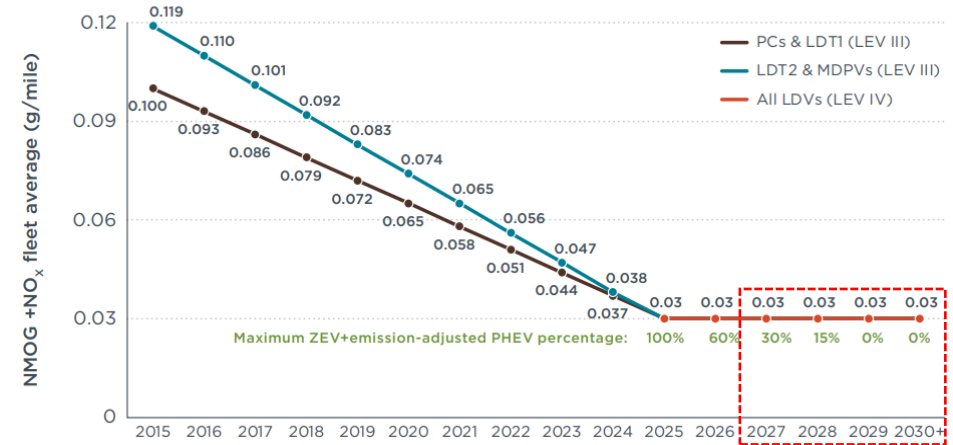


Figure 1. Fleet average NMOG+NOx standards for light-duty vehicles under California LEV III and LEV IV regulations

^a For each model year, a manufacturer may only include up to the specified percentage of its total ZEVs + emission-adjusted PHEVs in the fleet average calculation

^b Only applicable to manufacturers optionally certifying 2025 model year test groups





DURABILITY REQUIREMENTS

- ❖ The full useful life of vehicles certified to the **LEV IV** standards shall be **15 years or 150,000 miles**, whichever occurs first.
 - These full useful life values apply to all exhaust, evaporative, and refueling emission requirements except for standards which are specified to only be applicable at the time of certification.
- ❖ **California OBD**
 - All vehicles shall be subject to the provisions of title 13, CCR, section 1968, et seq., as applicable.
 - No vehicle shall be certified unless the Executive Officer finds that the vehicle complies with the requirements of title 13, CCR, section 1968, et seq., as applicable.
- ❖ **Vehicle Labeling**
 - This vehicle conforms to California regulations applicable to XXX-fueled 20XX model-year new (specify ULEV400, ULEV270, ULEV250, ULEV200, ULEV125, ULEV70, ULEV60, ULEV50, ULEV40, SULEV230, SULEV200, SULEV175, SULEV170, SULEV150, SULEV125, SULEV100, SULEV85, SULEV75, **SULEV30**, SULEV25, SULEV20, SULEV15, or ZEV, as applicable) (specify passenger cars, light-duty trucks, medium-duty vehicles).
 - For federally certified vehicles certified for sale in California the statement must include the phrase "conforms to U.S. EPA regulations and is certified for sale in California." Such statements shall not be used on labels placed on vehicles or engines which, in fact, do not comply with all applicable California regulations, including assembly-line test requirements, if any.

